

Code Layout, Readability and Reusability

Date:02 July 2026

Team ID:

Project Name: Smart Agricultural Production Optimization Engine

Maximum Marks: 5 Marks

S.No	Code Quality Parameter	Description	Followed (Yes/No/Partial)	Remarks
1.	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Consistent indentation maintained across all project files
2.	Proper File Structure Yes	Files and folders are logically organized	Yes	Project organized into Data Collection, Preprocessing, Model, Prediction, and Documentation modules.
3.	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables such as soil_type, rainfall, temperature, humidity, and crop_prediction are descriptive.
4.	Function / Method Names	Functions are descriptively named	Yes	Functions like collect_data(), preprocess_data(), train_model(), and predict_crop() are clearly named.
5.	Code Comments	Inline and block comments explain logic	Yes	Comments are provided for preprocessing, model training, and prediction logic
6.	Modular Design	Code is split into	Yes	Separate

		reusable functions/modules		modules are created for data processing, prediction, and user interface.
7.	No Redundant Code	Duplicate or unused code is removed	Yes	Duplicate code has been removed and reusable functions are used.
8.	Error Handling	Exceptions and errors are handled gracefully	Yes	Input validation and exception handling are implemented to improve reliability.

Reusable Components / Modules

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1.	Data Preprocessing Module	Python, Pandas	Data Cleaning	High
2.	Crop Recommendation Module	Python, Scikit-learn	Prediction Engine	High
3.	Soil Analysis Module	Python	Recommendation System	High
4.	Weather Analysis Module	Python	Crop Prediction	Medium
5.	User Interface	HTML, CSS, JavaScript	Frontend	High

Overall Code Quality Assessment

Aspect	Rating (1–5)	Comments
Code Layout &	5	Well-organized project structure.

Structure		
Readability	5	Code is easy to understand and maintain.
Reusability	5	Modules are reusable across the project
Documentation / Comments	5	Sufficient comments and documentation are provided.
Overall Score	5	Excellent code quality